

When Roads Close, Pollution Moves: Agent-Based Modeling of Street-Level NO₂ and Noise Redistribution in Portland

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Abstract Most street-level traffic-pollution maps use static land-use models. Because their inputs do not change with traffic, these maps cannot show what happens when drivers reroute. This study developed an agent-based model of vehicles moving through Portland's street network, including car-following, traffic signals, queues, and rerouting. The simulated traffic was then used to create street-segment NO₂ and noise surfaces. A road-closure experiment compared the agent-based results with a well-performing static land-use surface under the same closure. The static surface stayed unchanged because of how it was built. The agent-based model showed modeled NO₂ falling by about 65% along the SE Powell corridor and rising on parallel routes. This pattern held across 12 random seeds and tests on three arterials. In a separate comparison using 332 measured NO₂ sites, adding the model's traffic features to a fair land-use random forest improved summer prediction by 0.06 R^2 but did not improve winter prediction. The closure also lowered modeled noise on Powell by 3.8 dB while shifting some noise to alternate routes. Because Portland does not have dense monitoring, these are model-to-model comparisons. The study shows what traffic interaction modeling can add, but it does not claim absolute accuracy.

1 Introduction

Street-level traffic-pollution maps show where pollution is likely to be higher or lower across a city. They can support planning, route choices, decisions about where to place buildings or services, and studies of who may face greater exposure. This matters because traffic pollution can change sharply from one street to the next, especially near busy roads, intersections, traffic signals, and detour routes.

Many detailed pollution maps use static models, such as land-use regression or land-use random forests [5, 12]. These models rely on features that mostly stay fixed,

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including land use, road type, population density, and distance to major roads. They work well for estimating normal or average conditions. The problem comes when traffic changes. If a street closes, the cars do not disappear. Drivers use other streets, queues form in new places, and nearby roads may carry more traffic. A static model cannot directly show that movement because its inputs do not change with current traffic. This does not make static models bad. It means they answer a different question: they estimate pollution from fixed city features rather than showing how pollution shifts after traffic is rerouted.

This chapter tests what changes when the traffic itself is modeled. I built an agent-based model of vehicles moving through Portland’s street network. Vehicles follow one another, stop and queue at signals, slow in congestion, and take different routes through the network. I use the simulated traffic to create street-level NO_2 and traffic-noise surfaces. The key difference is that these surfaces are tied to what traffic does in the simulation, not only to fixed features on a map.

The main experiment is a road-closure test. I compare a static pollution surface with an agent-based surface under the same closure. The static surface stays the same because its inputs stay the same. The agent-based surface changes because vehicles reroute, creating new traffic and modeled NO_2 patterns. This study does not claim that the agent-based model is automatically more accurate than an existing land-use model. Portland has limited NO_2 monitoring data, so the comparison focuses on what each approach can represent. The claim is specific: a source-based traffic model can show pollution shifting after a closure, while a static model cannot show that shift because of how it is built.

I also test whether traffic predictors from the agent-based model add information to a random forest based on a published Portland study [12]. Both comparisons use the same type of statistical model, so the main difference is the source of the predictors. This makes it possible to test what simulated traffic adds beyond fixed geographic features.

2 Related Work and Background

Street-level traffic-pollution maps often use static methods such as land-use regression and land-use random forests. These models estimate pollution from fixed city features, including land use, road type, population density, and distance to major roads. Hoek et al. [5] review this family of methods. Rao et al. [12] provide the Portland example used to guide this project. Their work supplies the multi-buffer predictor design and the measured NO_2 sites used in Section 4.3. The closure experiment does not use Rao’s fitted model or measured data. It uses a Rao-style baseline built for this project. Static land-use modeling is not treated as a weak comparison here. It is the standard approach against which this project is tested.

Other studies model the traffic source more directly. They simulate traffic activity and then use that activity to estimate emissions and noise instead of relying only on fixed map features. Tirico et al. [17] are the closest prior work because they connect

traffic modeling to both NO_2 and CNOSSOS-style road-traffic noise. The main difference is the research question. Tirico et al. compare management scenarios within one dynamic modeling system. This chapter compares a source-based agent model with a static land-use surface. It removes road segments to test whether the models can show traffic pollution moving onto detour routes. That direct comparison is the gap addressed here.

Dynamic traffic-noise modeling with CNOSSOS is already an established area of research. Le Bescond et al. [7] created an open-source system that links dynamic traffic simulation with noise modeling. Baclet et al. [1] used calibrated microscopic traffic simulation to produce near-real-time noise maps. Pascale et al. [11] directly compared a fully dynamic microscopic method with static CNOSSOS-EU, so this chapter does not claim to be the first dynamic CNOSSOS study. Galassi Luquezi et al. [4] describe remaining limits in agent-based transport models for noise exposure, and Sharma et al. [14] provide an example of a static statistical noise map built from geographic predictors. The noise contribution in this chapter is narrower: it creates a source-based surface and tests whether modeled noise shifts when traffic is rerouted. A static noise map built from fixed geographic inputs cannot show that response.

Traffic models are also commonly checked against observed traffic counts. Road-network structure can explain part of the traffic pattern, but actual travel demand matters too. McDonald [8] separates these effects by comparing structure-based and demand-aware traffic prediction. I use that distinction to interpret the traffic-count results in Section 4.1. The goal is not to claim that the model exactly matches real traffic volumes. The goal is to check whether it places busy and quiet streets in a reasonable order before using it for the closure experiment.

The model uses established methods for its main technical parts. Vehicle movement uses the Intelligent Driver Model for car-following [18]. NO_x emissions are estimated from vehicle speed and acceleration with HBEFA-based factors [6]. Road-traffic noise is calculated with CNOSSOS, and the FHWA Traffic Noise Model provides a comparison point [3, 9]. The evaluation also uses spatial block cross-validation because nearby locations are likely to be similar and should not be treated as fully independent [13].

3 Data and Methods

I built the model in layers. The first layer is the street network. The second controls vehicle movement and car-following. The third adds traffic signals and queues. The remaining layers add trip demand, routing, emissions, noise, and the traffic predictors used in the forest comparison. This section explains how the model works rather than reporting its results. The main study area is the SE Powell Boulevard corridor in Portland, within a 1.5 km radius of Cleveland High School. This corridor model is a proof of concept and a fallback scale, not a model of the entire city. The main run uses a fixed random seed so it can be repeated. The simulation and figure-making steps

are also kept separate, so figures are created from saved results instead of rerunning the model.

The street network comes from OpenStreetMap through OSMnx [2]. The model therefore uses a real street network rather than a simplified grid. The Powell corridor graph contains 978 nodes, 2,838 directed edges, and 21 intersections tagged with traffic signals. Each direction of a street is represented as a separate segment. In the current model, each directed segment acts as one following lane. This limits road capacity, so I return to it in the limitations section. Figure 1 shows the corridor network.

Vehicle movement uses the Intelligent Driver Model [18]. Each vehicle tries to reach its target speed on an open road but slows when it gets too close to the vehicle ahead. The safe following distance grows with speed, so vehicles leave more space when moving faster and close the gap when traffic slows. This is one reason to use an agent-based model: the traffic pattern develops from many vehicles reacting to one another rather than being assigned in advance. The model also records each vehicle's speed and acceleration at every step, which are later used to estimate emissions.

Traffic signals and queues are added to the car-following model. At a red light, the model places an imaginary stopped vehicle at the stop line. This allows the same braking rule to handle both following another vehicle and stopping at a signal. When the light turns green, the imaginary vehicle is removed and traffic can move again. The model also lets a queue on the next street segment affect vehicles upstream. If the next segment is backed up, a vehicle waits at the stop line instead of entering the intersection and overlapping with the queue. Signal timing is simplified to a uniform 60-second cycle with 50 percent green time. This is an assumption, not Powell's actual signal plan. Powell uses adaptive SCATS control [16], which is discussed in Section 5.1.

Vehicles choose routes by travel time rather than distance because drivers generally care more about how long a trip takes than how short it is on a map. A gravity-style demand model creates origins and destinations using population, jobs, and distance. Demand decreases over a 1.5 km distance scale. The corridor run uses 500 vehicles, and 30 percent are through-traffic. These vehicles enter through one boundary of the study area and leave through another instead of starting and ending inside the corridor. I chose the vehicle count, distance scale, and through-traffic share before comparing the model with the held-out PBOT counts. This keeps the traffic-count test in Section 4.1 independent rather than tuning the model to the test data.

The emissions layer uses each vehicle's speed and acceleration to estimate NO_x . The original corridor runs use the HBEFA3 passenger-car diesel class PC_D_EU4, and emissions are added to the street segments where the vehicles travel [6]. The model then estimates NO_2 as a fixed 30 percent of NO_x : $\text{NO}_2 = 0.30 \times \text{NO}_x$. This conversion happens after the traffic simulation, so the fraction can be changed without rerunning traffic. Vehicle interaction matters because idling and stop-and-go driving emit more than steady cruising. In this emissions function, idle NO_x is about 13 mg/s, compared with about 6 mg/s during reference cruising. The original all-diesel surface should therefore be treated as an upper-bound estimate. A mixed-fleet check is reported later in the chapter.

The same simulated traffic also creates a traffic-noise surface. For each street segment, the model estimates sound power with the CNOSSOS method for category-1 vehicles [3]. It calculates tire and engine noise across octave bands, adjusts the bands for human hearing, and uses a simple line-source calculation to estimate noise beside the road. Because it uses the speed produced by the simulation, the noise layer responds to congestion rather than relying only on the posted speed. I checked the category-1 coefficients, reference speed, band order, and A-weighting against Directive (EU) 2015/996 Appendix F Table F-1. This first version models cars only and leaves out ground effects, barriers, buildings, and weather. These limits are discussed again in Section 5.1. The FHWA Traffic Noise Model is the planned comparison point for this surface [9].

The model also turns its traffic output into predictors for the later random-forest comparison. To match Rao’s design, traffic activity is summarized around each segment at five buffer distances from 100 m to 1,200 m [12]. The feature-building code uses an explicit list of allowed inputs. The agent-based model’s own NO_2 result is excluded, so the answer cannot leak into the predictors used to evaluate it. The forest comparison, spatial cross-validation, and R^2 results are reported in Section 4.3.

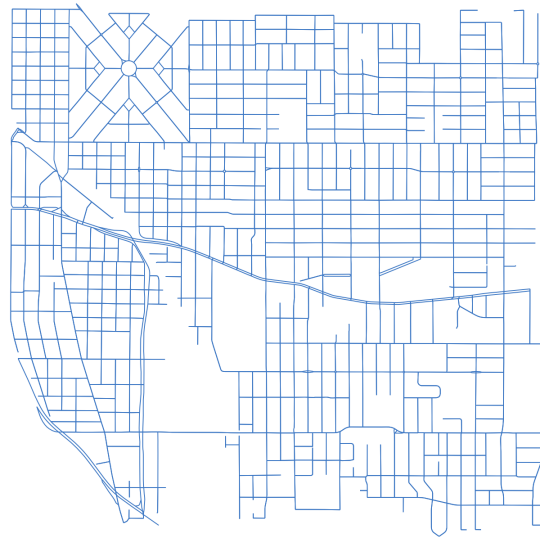


Fig. 1 The Powell corridor street network used for the corridor simulation (978 nodes, 2,838 directed edges; OpenStreetMap via OSMnx).

4 Results

4.1 *Validating the Traffic Layer*

Before using the model for the road-closure experiment, I checked the traffic layer in three ways. First, I tested whether individual vehicles followed the model's equations and physical rules. Second, I compared the modeled street pattern with traffic counts that were not used to build the model. Third, I checked whether a full-day run followed the intended rise and fall in daily demand. Together, these tests cover vehicle behavior, the ordering of busy and quiet streets, and the model's response to changing demand.

The first check used four small tests with results that could be calculated or inspected directly. In the car-following test, the simulated gap settled at 9.58 m, compared with 9.5 m from the IDM formula calculated by hand. In the signal test, the vehicle stopped exactly 2 m before the red-light stop line, matching the model's minimum stopping gap. In the saturation test, raising demand to 1,500 vehicles lowered average speed from 9.58 to 3.50 m/s and left about 34 percent of vehicles stopped. This showed that congestion developed from vehicle interaction rather than from a separate congestion rule. In the dense spillback test, the maximum distance past a segment endpoint was 0.0000 m. These tests confirmed that following, stopping, congestion, and segment boundaries worked as intended before the full corridor run.

The network-wide check used traffic counts that were held out from model construction. The observed data are average daily traffic counts, while the simulation counts vehicles passing a segment during one modeled hour and does not yet have a calibrated citywide demand total. The two sets of numbers are therefore not on the same scale, so exact numerical agreement would not be a useful test. Instead, I use Spearman rank correlation to ask whether the model and observed data place the same streets near the busy and quiet ends of the network. This tests the spatial traffic pattern without claiming that the model already reproduces exact traffic volumes.

With only gravity-based trips that began and ended inside the corridor, the model reached a Spearman correlation of 0.507 with the held-out PBOT counts. Adding the through-traffic component raised the correlation to 0.590. The through-traffic share, gravity distance, and vehicle count were based on network geometry and outside information rather than adjusted to raise this score. The improvement shows that trips crossing the corridor help the model rank streets more realistically while leaving the observed counts as an independent test.

The validation also uncovered an important matching error. An earlier version paired each observed count with the midpoint of a street. Near intersections, that method could connect a count to the wrong street, producing a correlation of 0.39. I changed the method to use the nearest point along the street geometry and reran the comparison. The correlation rose to 0.59 even though the traffic simulation had not changed. The gain came from fixing how the result was measured, not from changing the model. For that reason, I treat the earlier 0.39 value as an outdated result rather than a competing one.

To place the 0.590 result in context, I use McDonald’s distinction between structure-based and demand-aware traffic prediction [8]. In his San Francisco taxi data, methods based only on road structure reached about 0.29 to 0.31. Comparable results for Porto were about 0.25 to 0.40. Methods that included demand information generally reached about 0.7 to 0.9. My result falls between those ranges, which is reasonable for a model that includes road structure and partial demand information but not measured origin-destination demand. This is only a comparison for context, not a direct benchmark, because McDonald studied taxi volume in San Francisco and Porto rather than average daily traffic in Portland.

The 24-hour run provided a final consistency check. Network throughput followed the input demand profile with a Spearman correlation of 0.983. This is not independent validation because the daily profile is an input. It only confirms that the simulated traffic rises and falls in the expected direction. The run also shows how vehicle interaction affects emissions. With the approved mixed fleet, a vehicle traveling during the peak hour emitted about 5 percent more modeled NO_2 than a vehicle traveling off peak. Under the earlier all-diesel upper-bound assumption, the difference was 33 percent because diesel idling produces a larger congestion penalty. In both cases, the increase came from more slowing, stopping, and waiting. Vehicle interaction still matters, but the fleet mix changes the size of the effect.

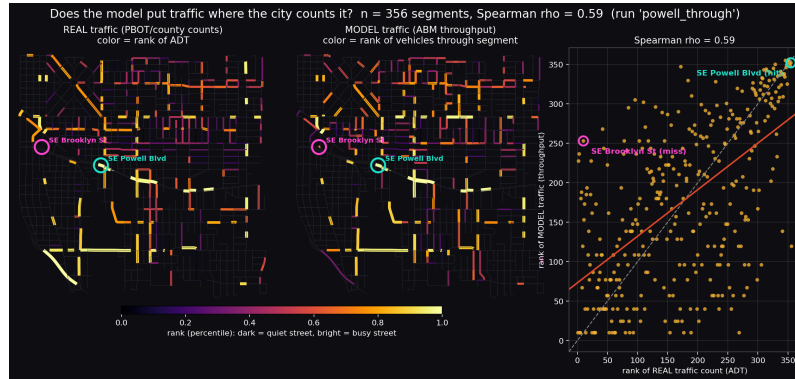


Fig. 2 Model traffic beside the held-out PBOT counts on the same streets (rank-colored), with the Spearman scatter.

4.2 The Road-Closure Experiment

The road-closure experiment is the main comparison between the static and agent-based surfaces. It asks a simple question: if part of a road is removed from the network, can the pollution surface respond? The static comparison is a 29-predictor Rao-style land-use random forest fitted to the agent-based model’s own open-scenario

NO₂ surface. It reached an out-of-bag R^2 of 0.714 on the log scale used for fitting. This gives the static model a fair and strong starting point because it is trained directly on the open agent-based surface. Even so, it cannot respond to the closure. The road closure does not change any of its geographic inputs, while vehicles in the agent-based model can reroute.

For the closure test, I removed a 150 m zone containing 24 directed street segments and reran the traffic model. Figure 3 shows the main result. The static surface shows no change, while the agent-based surface changes on the roads used for detours. In the representative seed-42 run, closing the SE Powell zone lowered corridor-wide modeled NO₂ on Powell by about 65 percent. At the same time, modeled NO₂ rose by about 41 percent on SE Division and about 57 percent on SE Holgate. The modeled pollution moved away from the closed corridor and onto the parallel roads that carried the rerouted traffic.

The changes on individual roads were large, but the change across the full network was much smaller. In the same seed-42 run, total modeled NO₂ rose by about 5 percent. This shows that the main effect was redistribution rather than a large overall increase or decrease. The closure changed where the modeled NO₂ appeared, which is the type of network response the agent-based model was built to represent.

To make sure this was not an unusual result from one random run, I repeated the experiment across 12 seeds and three arterial closure locations. Across the three arterials, the average drop on the closed corridor ranged from 61 to 76 percent. This is a range of the three arterial averages, not a range across individual seeds. The seed-42 result shows how the shift happens, while the repeated runs show that the same pattern holds across different seeds and closure locations.

I also combined the segment-level NO₂ changes with block-group population data to estimate how many residents live near streets where the modeled surface changes. In the corridor test, 11,154 residents are near segments where modeled NO₂ rises, and 12,394 are near segments where it falls, out of 23,548 residents in the affected block groups. These are modeled changes, not measured air-quality changes, personal exposure estimates, or health outcomes. The calculation shows why location matters: when traffic moves, the modeled burden also moves across neighborhoods and residents.

Overall, the closure experiment answers the chapter's main "what if" question. A static land-use surface can estimate a stable pollution pattern, but it cannot respond when the road network changes. The agent-based surface can respond because the traffic source moves. This does not show that the agent-based model is more accurate in absolute terms. It shows a narrower result: modeling traffic interaction can represent pollution shifting after a road closure, while a static model cannot represent that shift with unchanged inputs.

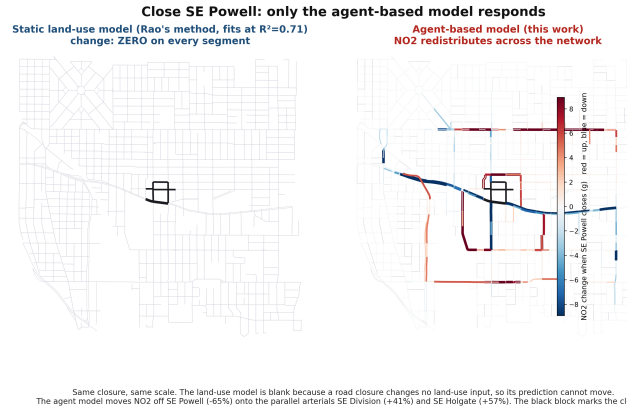


Fig. 3 The same road closure seen by both models, on one change scale. The static land-use surface cannot change (left); the agent-based surface redistributes modeled NO₂ onto detour routes (right).

4.3 The Model-to-Model NO₂ Comparison

This comparison asks whether simulated traffic adds useful information beyond standard land-use measures. Both models use the same random-forest method and the same measured NO₂ locations. One model uses only fixed land-use features, while the other adds traffic features from the agent-based model. The analysis requires the 20 km network because the Powell corridor contains only five measurement sites. The metro network includes 332 summer sites and 79 winter sites near the road network. The simulation uses LODES travel patterns from 531,245 commuters, 16,500 modeled vehicles, and a 15% through-traffic share chosen before testing.

Both sets of predictors were calculated within buffers from 100 to 1,200 meters. I tested performance with spatial block cross-validation, which keeps nearby monitoring sites together instead of placing them in both the training and test data [13]. I focus on the 2 km and 3 km blocks because 1 km blocks allow too much information to pass between nearby sites. A shuffle test checked whether the combined model improved only because it had more variables. When the traffic values were randomly moved to the wrong sites, performance dropped toward the land-use baseline. At 2 km, the shuffled model scored 0.388 ± 0.014 , compared with 0.460 for the real combined model. At 3 km, it scored 0.426 ± 0.012 , compared with 0.493. The improvement therefore depends on placing the correct traffic information at the correct locations.

An early version of the land-use model had two problems that made it weaker than intended. First, population and jobs were placed at block-group center points. As a result, 284 of the 332 sites had no population inside their 200 m buffers. Second, using only Oregon data left out 294,284 Clark County residents and 124,173 jobs in Washington. I corrected both problems by using Oregon and Washington data and spreading population and jobs across the areas where they are located. The land-use

model improved, but the benefit from adding simulated traffic stayed nearly the same. Correcting two problems that had favored the traffic model did not remove the result.

For summer, the corrected land-use model produced R^2 values of 0.437, 0.394, and 0.430 with 1, 2, and 3 km blocks. The combined model produced 0.493, 0.460, and 0.493. This was an improvement of about 0.06 at every block size. Winter was different. Adding traffic changed performance by only -0.012 to +0.006. One possible reason is that traffic has a stronger effect on summer NO_2 patterns, while heating and other residential sources matter more in winter and are better represented by population features. This is a possible explanation, not a tested conclusion. The results do not show that the traffic features generalize across seasons. Figure 4 shows that the real combined model exceeds the shuffled comparison range at all three block sizes.

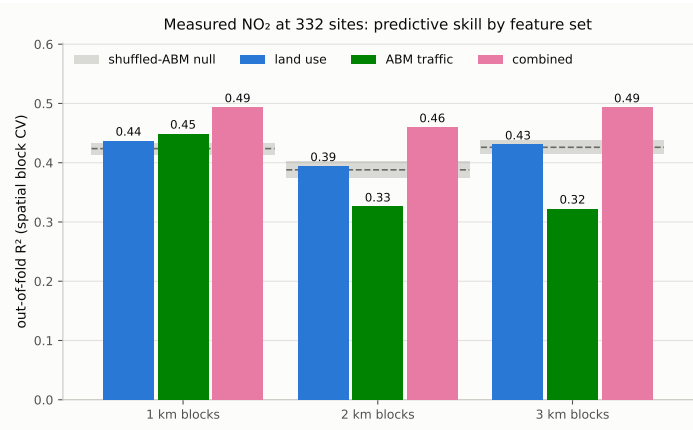


Fig. 4 Out-of-fold R^2 under spatial block cross-validation for the land-use, agent-based, and combined feature sets at three block sizes, with the shuffled-ABM null band. The combined model exceeds the null at every block size.

4.4 The Noise Surface

The simulation can also estimate traffic noise because it already records the number and speed of vehicles on each street segment. The same traffic results can create a noise surface without another traffic run. Unlike the NO_2 comparison, the noise calculation uses established sound equations rather than a model trained on monitoring data.

CNOSSOS separates vehicle noise into tire noise and engine noise. Tire noise increases with speed, while engine noise matters more at lower speeds. The model calculates sound across eight frequency bands, adjusts those bands for human hearing,

combines the sound along each road segment, and estimates the level at a receiver 10 meters away [3]. I checked the category-one coefficients, reference speed, band order, and hearing adjustments against the directive. A separate check produced about 96 dB(A) of sound power for one vehicle traveling at 50 km/h, which agrees with published CNOSSOS examples.

In the `powell_through` run, modeled segment noise ranged from 26.5 to 59.5 dB(A), with a median of 39.9 dB(A) across 2,838 street segments. Powell and Division were among the loudest roads, while streets with little or no traffic stayed quiet. The calculation responds to congestion because it uses each segment's simulated speed and traffic activity. Slow, congested traffic creates more engine noise. Changing from the all-diesel to the mixed-fleet emissions setting does not change the noise surface because vehicle speeds and flows stay the same and the current noise model treats every vehicle as a car.

The road closure shifted noise in much the same way that it shifted NO₂. Closing the Powell zone lowered modeled noise on Powell by about 3.8 dB(A), which represents more than a 50 percent drop in acoustic energy. Noise rose by up to about 1 dB on parallel routes, with the largest increases on quieter residential streets. The network-wide level changed by less than 1 dB. Across 12 seeds and all three closure locations, noise fell on the closed arterial in every run, with average drops of 3.5 to 7.0 dB depending on the location. Increases on parallel roads were smaller and varied more from seed to seed. A static noise map would stay unchanged because rerouting does not change its fixed inputs.

This first version leaves out several factors. It does not include trucks, barriers, buildings, ground absorption, reflections, or weather. Its simple propagation method produces values about 3 dB lower than the half-space convention, but that difference is consistent across segments. On heavily congested roads, the sound-source calculation also uses a minimum speed of 20 km/h even though the traffic-flow calculation keeps the actual speed. Because of these limits, the surface is meant for comparing relative street-level noise patterns, not for making an absolute environmental noise assessment.

Portland does not have enough street-level noise measurements to validate the full surface directly. I therefore use a model-to-model comparison similar to the NO₂ analysis. Shen et al. [15], for example, compared a CNOSSOS surface with other noise models and reported correlations from 0.39 to 0.77. The FHWA Traffic Noise Model is the most appropriate reference for this project [9]. A direct FHWA comparison is in progress and is the next planned step for the noise surface.



Fig. 5 The CNOSSOS street-segment noise surface for the corridor run: dB(A) at a 10 m receiver, brightest on Powell and Division.

5 Conclusion

5.1 Limitations and Honest Framing

This study shows what the model can represent. It does not claim that the agent-based model gives more accurate absolute pollution levels. Portland’s regulatory monitoring network is too sparse to validate a street-segment NO_2 surface by itself. The metropolitan area has only two regulatory NO_2 monitors [10]. Denser local data do exist through passive-sampler campaigns, including the measurements used by Rao et al. [12], and those data support the separate measurement-based comparison. The road-closure experiment should therefore be read as a test of how two model types respond to the same closure. It shows that the agent-based surface can respond to a road-network change while the static land-use surface stays fixed. It does not show that the modeled concentrations are the same as those that would be measured after a real closure.

The traffic model simplifies both signal timing and road capacity. SE Powell Boulevard uses adaptive SCATS signal control [16], but the model does not recreate Powell’s actual signal sequence. Instead, every modeled signal uses a 60-second, two-phase cycle with 50 percent green time. Because SCATS changes timing in response to traffic and no public fixed plan is available, this cycle is only an approximation of stopping and queue formation across the network. Each directed street segment also acts as one following lane, creating a capacity limit of about 1,070 vehicles per hour. Powell’s estimated directional peak is higher, at about 1,400 to 1,745 vehicles per hour. In addition, every vehicle uses the same car-following settings, so

the model includes vehicle interaction and queues but not differences among drivers in desired speed, reaction time, or aggressiveness. The peak-volume values are estimates rather than observations for every segment. These limits are why I judge the traffic model mainly by how it ranks streets rather than by whether it reproduces exact volumes.

The demand model is simplified too. In the corridor experiment, a gravity model creates trip origins and destinations with a 1.5 km distance scale, and 30 percent of vehicles are through-traffic. The model does not separate morning and evening trips by direction. I chose the vehicle count, distance scale, and through-traffic share from network geometry and outside information before comparing the model with the held-out PBOT counts. I did not change them to maximize the validation score. This keeps the traffic-count test from being tuned directly to its test data, but it is not a substitute for measured origin-destination demand.

The corridor simulation also has limited resolution on quiet streets. Throughput is counted in whole vehicles, so many low-traffic segments carry the same small number during a simulated hour and receive the same rank. This is a limit of the simulation size, not evidence of a coding error. The model separates busy arterials more clearly than it separates one quiet residential street from another. More vehicles or a longer observation period would create more differences among low-traffic streets. Spearman correlation handles tied values by assigning average ranks, but it cannot create differences that are not present in the simulation.

The original emissions surface should be treated as an upper-bound estimate rather than a calibrated estimate of real outdoor concentrations. The corridor runs use the HBEFA3 Euro 4 diesel passenger-car class, while Portland's real fleet includes different fuels, ages, and vehicle types. Recalculating emissions with the calibrated mixed fleet, while keeping the same traffic, lowered total NO_x by about 3.8 times at the corridor scale. The closure pattern still held: Powell fell about 63 percent, Division rose about 49 percent, Holgate rose about 59 percent, and the network total rose about 3 percent. The redistribution result therefore does not depend on the all-diesel assumption. The model also converts NO_x to NO_2 with a fixed fraction of 0.30, which is at the upper end of the literature range. These choices affect the total amount more than the open-versus-closed comparison because both scenarios use the same assumptions. The strongest result is the direction and location of the modeled change, not an exact concentration for a particular street.

Several code and reproducibility details also limit the model. Vehicle activity is assigned to the segment where a vehicle begins each simulation step, so very short links may receive slightly too little activity. The occupancy snapshot taken during each step can also briefly place two vehicles at the entrance of one segment. In addition, checkpoints do not save the random-number-generator state, so a resumed run will not exactly match an uninterrupted one. A completed checkpoint must also be cleared before the same run name is reused. These issues do not affect the reported closure pattern because the open and closed scenarios used the same activity rules and the closure sweeps did not resume from checkpoints.

The noise model has its own limits. On heavily congested segments, the CNOS-SOS sound-source calculation uses a minimum speed of 20 kilometers per hour,

while the traffic-flow calculation keeps the actual vehicle speed. The propagation calculation uses a cylindrical (2- π) field, which places the results about 3 dB below the half-space convention but does so consistently across segments. The model also leaves out ground absorption, barriers, building reflections, weather, and heavy-vehicle classes. The noise surface is therefore most useful for comparing relative patterns created by the simulated traffic. It is not a replacement for a complete environmental noise assessment.

The land-use comparison also uses some spatial approximations. An early version placed population and jobs at block-group center points, which left many monitoring-site buffers empty and could have favored the agent-based traffic features. The final comparison corrected this by spreading population and jobs across block-group areas and including both Oregon and Washington. The land-use model improved, but the added value of the traffic features remained. The site-centered pipeline also reduces a separate street-matching problem by sampling road geometry every 20 meters, although some demonstration paths still use street midpoints. Spatial block cross-validation reduces leakage between nearby sites, but two nearby points can still fall into different folds when they are on opposite sides of a block boundary. The results should therefore be read at the reported block sizes and with the effective number of folds.

Together, these limits show what the current results do and do not support. Modeling individual vehicles, queues, and route changes allows the street-level surface to respond to a road closure. Repeating the experiment across seeds shows that the redistribution pattern is not tied to one random run. The model does not yet predict the real NO_2 or noise level that would follow a specific Portland closure. A calibrated prediction would require a representative mixed fleet, measured origin-destination demand, more realistic lanes and signals, better environmental propagation, and a measurement campaign designed to test the model. The NO_2 -to- NO_x fraction, fleet mix, signal timing, and gravity distance are known calibration choices rather than hidden settings. Their final values will be selected with the project mentor from available evidence and documented criteria.

5.2 *Summary and Next Steps*

This study asked a direct question: Does modeling traffic as a moving and interacting pollution source add information beyond fixed land-use predictors? The answer was yes in some settings, but not all. In the road-closure experiments, modeled NO_2 and noise moved away from the closed road and onto alternate routes. The pattern held across random seeds and tests on three arterials. Adding the model's traffic features also improved summer NO_2 prediction compared with a fair land-use baseline. It did not improve winter prediction. The winter result matters because it shows that the value of dynamic traffic information may depend on the season or on other pollution sources.

This approach could help screen planning questions in which traffic patterns change, including road closures, construction detours, and new transportation infrastructure. Static pollution maps stay fixed when their inputs do not change. A source-based model can instead show where traffic-related pollution and noise may move, including whether they shift from major roads onto residential streets.

The model can be improved in several ways. One next step is to add CNOS-SOS categories for heavy vehicles and use vehicle-specific NO₂ fractions for the classes already assigned in the simulation. Another is to finish the comparison between the modeled noise surface and the Federal Highway Administration's Traffic Noise Model. Larger extensions could add cold-start emissions and atmospheric dispersion. A full test of absolute accuracy would still require a measurement campaign designed for this model, which was beyond the scope of a summer research project.

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Code availability.

The full simulation and analysis code is available at <https://github.com/darcy0408/portland-traffic-abm>. The repository includes the configuration file and the fixed random seed used for every result in this chapter. Dr. Meenakshi Rao shared the measured NO₂ site data for this comparison, so those data are not included in the repository.

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